

TABLE 1: The semi-structured interview questions.

Section 1: Background and Responsibilities
To get started, can you tell us a bit about your current role or position? How long have you been working in this field? What are your main responsibilities in your current role?
Section 2: CAV Challenges
Consider an incident where an autonomous vehicle encounters a roadblock, such as traffic cones, construction, or an irregular diversion of traffic. Have you ever come across such situations personally or otherwise? What are your thoughts on this scenario? Do you think it is plausible? What are the possible ways you think this situation will be handled, by both (1) the autonomous vehicle's design and (2) the company, if the autonomous vehicle is not designed to handle it?
Section 3: Teleoperation and Capabilities of Teleoperator
Are you familiar with teleoperation in connected & autonomous vehicles (CAVs)? Have you worked on teleoperation? In what capacity? Do you have experience using teleoperated vehicles or controlling a teleoperated vehicle? (a) Can you describe your experience? Are you aware of any scenario or new terrain that your CAV came across and found challenging to navigate? (a) Were there users in the vehicles? (b) How was the situation resolved? (c) What was the actual cause of this unusual behavior? Do you have any opinions and experiences involving a malicious teleoperator with ill intent? Do you think a teleoperator can hijack a CAV? What security and safety guarantees do passengers in teleoperated vehicles have while a remote operator controls the vehicle? How is a teleoperator service invoked or requested? (a) Is it an automated process, or do users have to request the service? Or is there a middleman or middleware that decides if teleoperator services are needed in specific scenarios? Should a teleoperator be able to monitor a CAV and have autonomy to decide if their services are required, especially in safety-critical situations? Do you think a teleoperator should have a direct line of communication with the passenger in a CAV? Should a teleoperator have access to all sensors/components onboard the vehicle? Which onboard components do you think a teleoperator should not be able to control or have access to? What are your expectations from a teleoperator in terms of qualifications and experience? Do you think they are functions a teleoperator should not be given access to? (a) Can you mention/describe these functions?
Section 4: Security and Privacy Challenges
Do you have any security concerns regarding teleoperation? To the best of your knowledge, are teleoperator identities revealed to the users of teleoperated vehicles? Do you think the teleoperator identities should be revealed to the passengers in CAVs? What aspects of teleoperation do you think are most vulnerable to security threats? Are you aware of any specific security or safety challenges that apply to teleoperated vehicles? (a) Are you aware of any defenses or techniques used to mitigate these challenges? Since cameras and other sensors are used in navigation, are there any privacy concerns users should be aware of in terms of recording their private conversations, which they may be having in the vehicles? What sensor data do you collect for your CAVs? What mechanism is used to transfer the data? How do you protect this data in transit from an eavesdropper? If data is not transferred; how is the data kept secure in the individual connected vehicles? What types of sensor data are processed in the CAV itself, and what types are also transmitted to remote services/cloud for processing? How frequently is the data transmitted? What happens to the sensor data after each trip? Who has access to this data? How long is it retained? Is it used for R&D?
Section 5: Security and Safety Processes in Design and Development
How relevant is the communication channel between a CAV and a teleoperator? Can you describe how a teleoperator maintains a constant connection for control of a CAV even in low bandwidth situations? Do you know some specific technologies or solutions to address the area where the signal is poor? Can you describe what you think is the system's architecture of teleoperation in connected & autonomous vehicles? When we want to design a teleoperation framework, what are the differences/challenges between traditional frameworks like cloud-edge server-vehicle? How important is security in the design and development process from your organization's perspective? Are there any safety & security training/workshops organized within the organization to sensitize engineers on organizational practices for safe and secure CAVs? What area of teleoperation design and implementation needs improvement? What are some of the things you will do differently? Have you worked on developing a teleoperated vehicle? Can you describe your development process from conceptualization to testing? Are there any testing mechanisms or procedures in place for trying out/testing new features after implementation? Do you run any specific tests for security? Are there other unusual scenarios you have encountered during testing or using CAV?
Section 6: Regulatory Guidelines and Legislation
Are you aware of any government regulations or laws that teleoperation companies/services have to comply with? Are there any organizational requirements or guidelines for safety and security? Are you aware of any institution that provides Teleoperation training or/and certification to individuals/organizations? What do you think about the current state of the regulations surrounding teleoperation and fully autonomous vehicles? Are they optimal, suboptimal or is there room for improvement? If there are security guidelines, have there been any times or instances where you have deviated from following these guidelines or have knowledge of a team or individual deviating from the guidelines? Are these guidelines based on universal standards or standards set by the organization? Do organizations have to comply with certain regulations or obtain certification before they can provide teleoperation services? What regulations do you think should be introduced? What are some of these guidelines as it relates to user safety and teleoperation? Are you aware of any guidelines or regulations that a teleoperator must follow while carrying out any service? Are there requirements that must be met before assuming the role of a teleoperator? Can you describe them, if any? Has there been any instance where due to pressure from consumers, organizations, or team members/lead, you sped up development & testing and completely or partially ignored security and safety concerns? In case of damages to the user or vehicles in a teleoperated CAV, who is liable financially for harm done or compensation as needed? In the case of insufficient regulations, if there is a violation of user privacy, who do you think should be responsible: the teleoperator, the organization, or the government?